MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPE 4MP1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	Meg	Vdc	Ratio
Maximum:	6.9	0	1,110	3,300	6,600	600	1.5	1.0 (see note 1)	±180	2.0 (see note 2)
Minimum:	5.7	200	---	1,000	1,000	---	---	---	---	---
TEST CONDITIONS: (see note 3)	6.3	Adjust	Focus	2,000	4,000	---	---	---	---	---

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MIN	MAX	
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	7	Ib3 = 25 µAdc	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5221	Light output	-	Ib3 = 25 µAdc	Light	25	---	ftL
5223	Modulation	-	Ib3 = 25 µAdc	ΔEc1	---	38	Volts
5231	Spot position (electrostatic deflection)			---	---	10	mm
5231	Spot displacement (leakage)			Displ	---	8	mm
---	Useful scan	-		---	2.875	---	In.
5241	Grid-cutoff voltage	-		Ec	-87	-52	Vdc
---	Trace distortion	4		---	---	0.02	In.
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
---	Pattern distortion	5		---	---	2	%
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mAdc
5201	Electrode current (anode No. 1)	-		Ib1	-15	+10	µAdc
5201	Electrode current (cathode)	-	Ib3 = 25 µAdc	Ik	---	500	µAdc
5101	Base alinement (electrostatic types)	-	+1D2, pin No. 5	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	+1D2	---	---	---	---
5101	Side terminal and base alinement	-	Pin No. 5	---	---	---	---
5101	Angle between traces	-		---	89	91	Degrees
5101	Trace and bulb alinement (rectangular electrostatic types)	-		---	---	1.5	Degrees

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued						
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5226	Line width "A" (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.45	mm
5226	Line width "B" (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.60	mm
5216	Cathode illumination	-		---	---	---	---
5246	Focusing voltage at cutoff	-		Eb1	---	510	Vdc
5246	Focusing voltage at modulation condition	-		Eb1	340	---	Vdc
5248	Deflection factor	-	1D2	DF	68	82	Vdc/ in.
5248	Deflection factor	-	3D4	DF	42	52	Vdc/ in.
5251	Heater cathode leakage current						
---	Grid emission	6		---	---	---	---
1101	Secureness of base, cap, or insert			---	---	---	---
(E)1111	Base pin solder depth	-		---	---	---	---
(E)1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Ib3 = 25 µAdc; t = 500 hours	---	---	---	---
---	Life test end points:						
5223	Modulation	-	Ib3 = 20 µAdc	ΔEc1	---	38	Vdc
5226	Line width "A"	-		Width	---	0.45	mm
5226	Line width "B"	-		Width	---	0.60	mm
	Periodic-check tests						
(E)1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Periodic-check tests -Continued						
5101	Neck and bulb alinement (electrostatic types)	-		Dia	---	1.87	In.
5248	Deflection-factor uniformity	-		---	---	2	%
1331	Direct-interelectrode capacitances						
	k to all	-		Ck	---	5.0	pF
	g1 to all	-		Cg1	---	6.2	pF
	D1 to D2	-		CD1	---	2.7	pF
	D3 to D4	-		CD3	---	1.9	pF
	D1 to all	-		CD1	---	5.8	pF
	D2 to all	-		CD2	---	5.8	pF
	D3 to all	-		CD3	---	4.3	pF
	D4 to all	-		CD4	---	4.3	pF
1141	Pressure	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
5216	Stray light emission (conventional types)	-	Eb2 = 3,300 Vdc Eb3 = 6,600 Vdc	---	---	---	---

NOTES:

1. It is recommended that the deflection electrode circuit resistances be approximately equal.
2. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity and pattern distortion.
3. All tests shall be performed with the tube shielded.
4. The focused sides of a raster pattern centered with respect to the tube face and whose mean dimensions are 2-1/4 by 2-1/4 inches (57.15 by 57.15 mm), shall not deviate from straight lines passing through the corners by more than the value specified herein.
5. All portions of a raster pattern, adjusted so its widest points just touch the sides of a 2.295-inch (58.29 mm) rectangle, will fall within the area bounded by the 2.295- by 2.295-inch (58.29 by 58.29 mm) rectangle and an inscribed 2.205- by 2.205-inch (56.01 by 56.01 mm) rectangle.
6. The tube shall be operated for a minimum of 5 minutes with Ef = 6.9 V, Eb2 = 3,300 V, Eb3 = 6,600 V. Ec1 shall be maximum negative; positioning and deflection voltages shall be zero. Viewed under dark conditions with dark-adapted (5 minutes) eye, there shall be no evidence of fluorescent light when Eb1 is varied from 0 to 1,110 volts. This test shall be performed immediately following the completion of the other electrical tests.
7. This test to be performed at the conclusion of the holding period.

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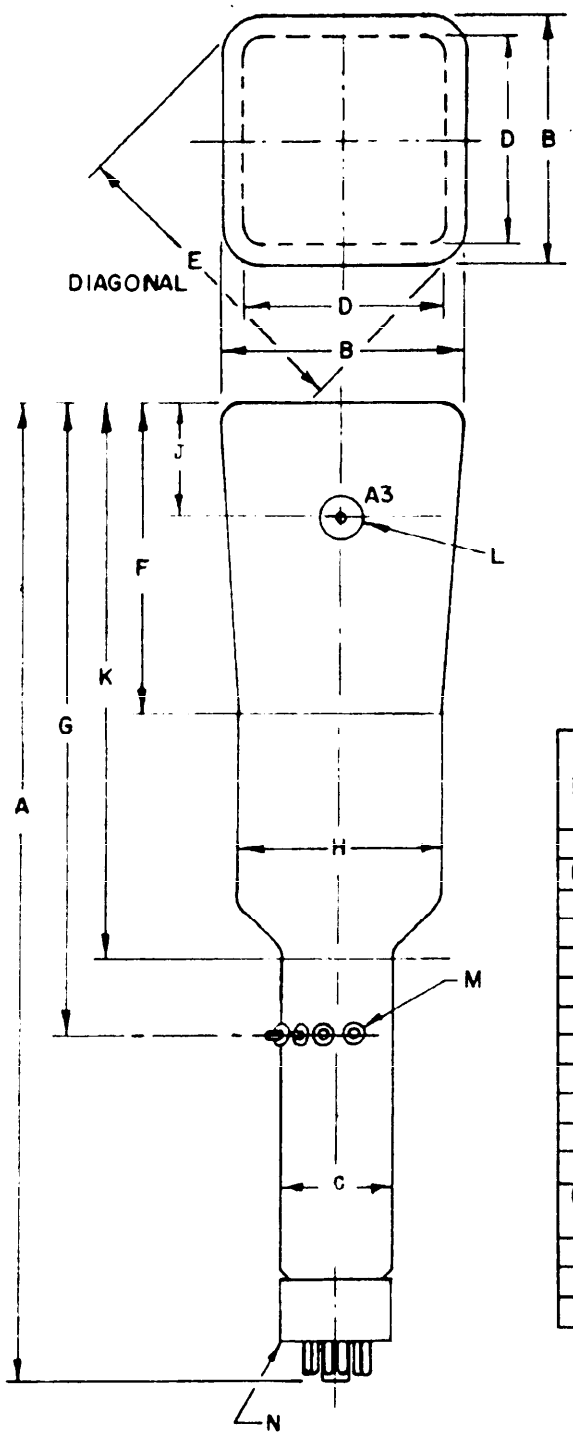
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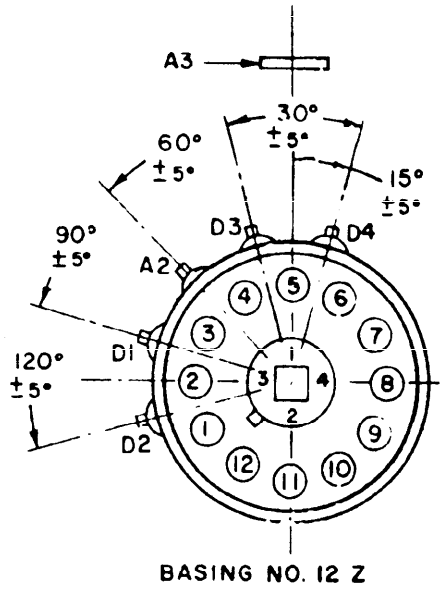
Preparing activity: Navy - EC

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BASING NO. 12 Z

NOTE:
+ ID2 TOWARDS PIN NO. 5

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	13.50 (342.90)	14.00 (355.60)
B	3.44 (77.32)	3.56 (90.42)
C	1.56 (39.62)	1.69 (43.15)
D	2.88 (73.15)	
E	4.16 (105.66)	4.34 (110.24)
F	4.13 (104.96)	4.88 (123.95)
G	9.00 (228.60)	9.25 (234.95)
H	2.97 (75.44)	3.03 (76.96)
J	1.50 (38.10)	1.75 (44.45)
K	7.63 (193.80)	7.88 (200.15)
Quality conformance inspection, part 3 (periodic check)		
L	J1-22 Contact	
M	J1-25 Contacts	
N	Base: B12-43	

PIN CONNECTIONS	
Pin No.	Element
1	h
2	g1
3	k
4	a1
5	ic
6	nc
7	nc
8	nc
9	nc
10	nc
11	nc
12	h

NOTES:
a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type 4MP1.

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